(3) There is only one absolute maximum (and one absolute minimum) but there can be more than one local maximum or minimum.

Points of Inflection

Definition 7.9

The points where the graph of the function changes from **"concave up to concave down"** or **"concave down to concave up"** are called the **points of inflection of** $f(x)$.

Theorem 7.12 (Test for Points of Inflection)

(i) If $f''(c)$ exists and $f''(c)$ changes sign when passing through $x=c$, then the point $(c, f(c))$ is a point of inflection of the graph of f.

(ii) If $f''(c)$ exists at the point of inflection, then $f''(c)=0$.

Remark

To determine the position of points of inflexion on the curve $y=f(x)$ it is necessary to find the points where $f''(x)$ changes sign. For 'smooth' curves (no sharp corners), this may happen when either

(i) $f''(x)=0$ or

(ii) $f''(x)$ does not exist at the point.

Remark

(1) It is also possible that $f''(c)$ may not exist, but $(c, f(c))$ could be a point of inflection. For instance, $f(x)=x^{\frac{1}{3}}$ at $c=0$.

(2) It is possible that $f''(c)=0$ at a point but $(c, f(c))$ need not be a point of inflection. For instance, $f(x)=x^4$ at $c=0$.

(3) A point of inflection need not be a stationary point. For instance, if $f(x)=\sin x$ then, $f'(x)=\cos x$ and $f''(x)=-\sin x$ and hence $(\pi, 0)$ is a point of inflection but not a stationary point for $f(x)$.

Example 7.57

Determine the intervals of concavity of the curve $f(x)=(x-1)^3\cdot(x-5), x\in\mathbb{R}$ and, points of inflection if any.

Solution

The given function is a polynomial of degree 4. Now,

$$\begin{aligned} f'(x) &= (x-1)^3+3(x-1)^2\cdot(x-5) \\ &= 4(x-1)^2\cdot(x-4) \\ f''(x) &= 4((x-1)^2+2(x-1)\cdot(x-4)) \\ &= 12(x-1)\cdot(x-3) \end{aligned}$$

Now,

$$f''(x) = 0 \Rightarrow x=1, x=3.$$

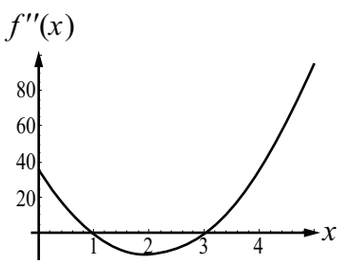


Fig.7.24

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